A Responsible Generative AI Approach to AI-driven Medical Diagnostic Systems Based on Design Thinking Principles

First A. Kajal Bhapkar

Abstract—**This report outlines the design and development of a medical diagnostic tool intended to accurately forecast diseases based on patient symptoms. Utilizing a dataset containing 132 symptoms linked to 42 disease outcomes, this initiative employs both conventional machine learning techniques and Design Thinking frameworks, enhanced by Generative AI, to produce a dependable, user-focused diagnostic solution. The primary models implemented—HistGradientBoostingClassifier and Random Forest Classifier—were refined through a process of continuous feedback and exploratory data analysis supported by AI-driven insights, culminating in a robust predictive system with significant potential for application in healthcare.**

I. Introduction

Assumption

Our main hypothesis was that a machine learning model developed on symptom-based datasets could attain high accuracy in predicting diseases. We investigated if the incorporation of Generative AI tools could improve data accuracy and diversity, while the Design Thinking method would refine the model's relevance for healthcare practitioners.

Inspiration for Generative AI and Design Thinking

As the need for quick and precise diagnoses in healthcare grows, utilizing AI to analyze symptoms offers significant possibilities. Generative AI improves this by elevating data quality via anomaly detection and simulation, thereby enhancing the model's flexibility with various patient data. At the same time, Design Thinking grounds the process in practical usability, guaranteeing that the AI solution meets the requirements of both physicians and patients.

The focal aim of this project is to create a disease classification model with machine learning to aid healthcare professionals in identifying diseases. This method integrates HistGradientBoosting with Random Forest algorithms, assessed for their precision and understandability. To improve the project's significance, Generative AI tools were employed within a Design Thinking framework to integrate user feedback at every stage, ensuring the system was both precise and user-friendly while upholding ethical standards.

II. Dataset Description

The dataset comprises of 2 CSV files, one for training purpose and the other for testing and validating, each file

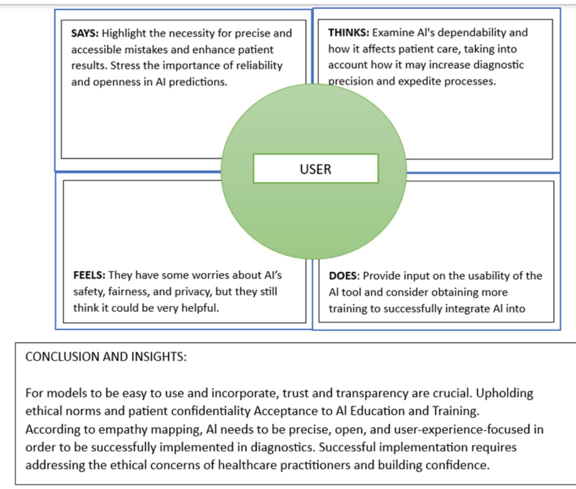

Fig. 1. Empathy Map

has 133 columns, out of which 132 have been used to depict the symptoms while one column describes the outcome of the disease. Generative AI techniques in data preparation have been used to expand the dataset and automate error detection, making the model better at predictions. The first phase of the project was focusing on grasping the requirements of end-users, especially physicians and patients. Empathetic methods were utilized to uncover their specific needs, allowing for a more user-focused approach to ideation around data. Collaborative discussions took place with related people to explore different data sources, each providing distinct diagnostic insights. The dataset was transformed into a format suitable for training by handling unnecessary columns , null values categorical variables and scaling numeric features. Overall, the data preparation stage set a solid foundation for accurate model training and evaluation.

III. Design Thinking Stages

A. Empathize

In an effort to create a user-centered diagnostic system, we first looked at the challenges faced by end users, or medical professionals. We created scenarios using generative artificial intelligence to record patient viewpoints, typical diagnostic trends, and the essential needs of healthcare providers. The decision to use of interpretability and precision criteria were impacted by these findings.

B. Define

After evaluating the needs of end users, we determined the primary goal: to create a system that combines remarkable diagnostic accuracy with user-friendliness for medical professionals. In order to improve the model selection process and highlight accuracy and clarity, generative AI helped analyze the relationship between symptoms and illnesses and highlight the key symptoms for diagnosis.

C. Ideate

During this stage, advanced classifiers appropriate for the complex nature of the data were recommended, model prototypes were created, and various classification algorithms were tested using generative AI tools. HistGradientBoostingClassifier and Random Forest algorithm were picked from among the generated options due to their interpretability and proved effectiveness in multi-class classification problems.

D. Prototype

1 Model Selection and Tuning

1) **HistGradientBoostingClassifier**: This technique is well-known for managing intricate data interactions and performs well in multi-class classification using sizable feature sets. Hyperparameter tweaking was carried out using generative AI techniques, which optimized learning rate, tree depth, and other parameters. 2) **Random Forest Classifier**: This model uses ensemble learning to increase classification accuracy and is both interpretable and resilient. In order to control overfitting while maintaining interpretability, AI-powered tuning increased the number of trees and maximum depth.

2 Data Preprocessing

1) **Missing Data Handling**: In an effort to maintain data integrity, generative AI-assisted approaches identified patterns in missing values and suggested imputation techniques. 2) **Feature Scaling and Encoding**: In order to guarantee that every feature was compatible with the model and prevent data distortion during training, AI suggested the best scaling and encoding techniques. AI recommended optimal scaling and encoding methods to ensure all features were compatible with the model, avoiding any data distortion during training.

E. Test

- Recall, accuracy, precision, and F1-score were utilized to assess performance. To present each model's predictions across disease categories, confusion matrices were also created.

  Generative AI in Testing: The AI simulated validation scenarios, tested both models on different patient demographics, and helped validate predictions under diverse conditions to ensure the models generalized well.
- **Key Results**:
  - **HistGradientBoostingClassifier** achieved 97.62% accuracy, with high recall and precision on major diseases.
  - **Random Forest Classifier** reached 97.62% accuracy, providing higher interpretability and balanced performance across disease classes.
  - **Results and Discussion** :
    The results demonstrated that both HistGradientBoosting and Random Forest classifiers performed well, with distinct strengths:
    * **Accuracy**: HistGradientBoostingClassifier showed slightly higher accuracy, especially in handling complex symptom patterns. The ensemble nature of this model, coupled with gradient boosting, allowed for nuanced decision boundaries.
    * **Interpretability**: Random Forest offered greater transparency with feature importance scores, a key consideration for healthcare providers seeking clear reasoning behind predictions. This model was particularly effective in diseases where the symptom-disease relationships were straightforward.
    * **Generative AI Impact**: The project's efficiency was increased by using generative AI, particularly during the preprocessing, model tweaking, and testing stages. Additionally, it facilitated rapid validation iterations, enhancing the resilience of the model.

    In contrast, Random Forest's transparent structure offered an advantage in user trust, making it more appropriate for situations where interpretability is crucial, whereas HistGradientBoostingClassifier's greater complexity came at the expense of interpretability.

IV. Ethical Considerations and Responsible AI Use

1 Addressing Bias

The distribution of symptoms among diseases was examined in order to check the dataset for potential biases. By detecting symptoms that disproportionately affect particular groups, generative AI systems helped discover bias. To reduce bias in model predictions, modifications were made to the data sampling procedure to guarantee fair representation across disorders.

2 Privacy and Transparency

Given that Generative AI played a significant role, maintaining data privacy was crucial. Furthermore, Random Forest's feature importance scores allowed medical practitioners to understand how predictions were generated, enhancing transparency.

3 Explainability
The unique model structure of HistGradientBoosting needed further explainability work. In order to visualize the value of features in decision-making and provide an enhanced understanding of model outputs.

V. Conclusion and Future Work

In order to illustrate the benefits of both methods for medical diagnostics, this study created a disease prediction model using the Random Forest and HistGradientBoosting classifiers. Data processing, model selection, and testing were all greatly impacted by generative AI, which made it easier to align the project with healthcare standards.

Future Work:

1) **Model Expansion**: In particular, for uncommon diseases, deep learning models may be investigated to capture intricate disease patterns.
2) **Real-Time Diagnostics**: By using Generative AI to simulate different patient demographics, this model might be integrated into a real-time diagnostic tool, thereby increasing accessibility.
3) **Enhanced Interpretability**: Developing fresh techniques for real-time interpretation of complex models, like HistGradientBoosting, could help bridge the accuracy-interpretability gap.

In conclusion, by striking a balance between technical accuracy, user needs, and ethical standards, this study demonstrates a potential future for AI-driven medical diagnostics.

VI. References

1) Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning.* MIT Press.
2) Shorten, C., & Khoshgoftaar, T. M. (2019). A survey on image data augmentation for deep learning. *Journal of Big Data, 6*(1), 60.
3) Aggarwal, C. C. (2013). *Outlier analysis.* Springer.
4) Tan, P. N., Steinbach, M., & Kumar, V. (2019). *Introduction to data mining* (2nd ed.). Pearson.
5) Géron, A. (2019). *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow* (2nd ed.). O'Reilly Media.
6) Hawkins, D. M. (1980). *Identification of outliers.* Chapman and Hall.
   This classic text provides foundational methods for outlier (anomaly) detection, which remains relevant in automated, AI-enhanced detection tools today.
7) Zimek, A., & Schubert, E. (2017). Outlier detection. In Liu, L., & Özsu, M. T. (Eds.), *Encyclopedia of database systems* (pp. 1-5). Springer.
8) Bengio, Y., Courville, A., & Vincent, P. (2013). Representation learning: A review and new perspectives. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 35*(8), 1798-1828.
9) Bishop, C. M. (2006). *Pattern recognition and machine learning.* Springer.
10) Hinton, G. E., & Salakhutdinov, R. R. (2006). Reducing the dimensionality of data with neural networks. *Science, 313*(5786), 504-507.